

Written Assignment 5- Intermediate code optimization

1. Please construct the DAG for the following basic block, and we assume that only variable “M” is live on exit, please optimize the block and rewrite the block in optimized code form

E=2+3

F=E-C

G=F*F

H=E-C

I=H-G

J=I+G

M=I[J]

2. Please construct the DAG for the following basic block, optimize the block and rewrite the block in optimized code form. Note that we assume only Variable G,L,M would be used later

B=3

D=A+C

E=A*C

F=D+E

G=B*F

H=A+C

I=A*C

J=H+I

K=B*5

L=K+J

M=L

3. Please construct the DAG for the following basic block, and we assume that only variable “M” would be used later, please optimize the block and rewrite the block in optimized code form.

T1=1*20

T2=T1+1

T3=T2*4

```

T4=addrA-T3
T5=i*20
T6=T5+j
T7=T6*2
T8=1*20
T9=T8+1
T10=T9*4
T11=addrB-T10
T12=i*20
T13=T12+j
T14=T13*2
T15=T11[T14]
A=3
T16=T15*A
T4[T7]=T16
M=T4[T7]
M=M*M
    L=M

```

Complete the homework and submit it before the next next class to the teaching assistant (Room 156, Building of Sch. Computer Sci&Eng).

1. Everyone should give their completed workbook to the teaching assistant before the designated time. There is a box at the door of the room specifically for collecting homework.
2. After the first submission of each assignment, it will be graded based on the quality of completion. Allow the next submission to modify errors from the previous assignment. The final score for each assignment will be given based on the final quality and number of revisions made.